Which reverts count? Reverter populations, attribution rules and observation windows behind a Wikipedia indicator of information quality work

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Abstract

Whether an edit was reverted is a standard trace of information quality work in Wikipedia, read as quality judgment, conflict, rejection and human effort. It depends on seldom-reported choices: who counts as a reverter, how a later revision is attributed to an earlier edit, and how long edits are observed. Of 41 studies, coded with language-model assistance and author checks, 22 do not indicate how bots are handled and 28 give no observation window, even counting choices recoverable from citations, defaults or released code; none varies the reverter population. The choices are measured on the first edits of 3.2 million newcomers in seven editions. Bots, one-click tools, other editors and the newcomers each make an edition-specific share of first reverts, and those reverted by different classes differ in later activity. For newcomer first edits, MediaWiki's revert tag omits most reverts by editors without patrol-exempt rights in the two editions that patrol recent changes: the gap between English and Italian revert rates is 7.4 points with the tag against 3.6 with a hash-based rule. Whether a filter's effect fades over longer windows depends on the attribution rule. Registered tests concern a secondary question; the main analyses are exploratory. Reporting items follow.

Keywords: information quality; trace data; measurement validity; Wikipedia; reverts; bots; operationalization

Introduction

When Stvilia et al. (2007) proposed a framework for information quality assessment in this journal, two of the measures they derived for Wikipedia articles were built from reverts. The number of reverts entered the measure of authority and reputation, and the median time to revert defined volatility. The article does not say how a revert was identified or whose reverts were counted. Nearly two decades later the revert is a standard trace of what Stvilia et al. (2008) called

information quality work in Wikipedia, and the omission is common: of 41 studies reviewed below, none states all of the choices on which a revert count depends.

Because every revert is recorded, “was this edit reverted” serves as a general-purpose indicator. It is read as a judgment on the quality of an edit and as a label for training quality-prediction models (Trokhymovych et al., 2023), as a measure of conflict (Kittur et al., 2007; Yasseri et al., 2012), as the rejection a newcomer experiences (Halfaker et al., 2011, 2013), as an outcome in onboarding experiments (Warncke-Wang et al., 2023) and as evidence about how humans and automated agents share quality work (Geiger & Halfaker, 2013; Zheng et al., 2019). These uses presuppose different constructs, and information science has long held that what an indicator of quality means depends on the activity and context in which it is used (Stvilia et al., 2007; Mai, 2013). A judgment on an edit, a dispute, an experience of rejection and a quantity of human effort are not carried by the same reverts.

A revert is a relation between an edit and a later revision, so the analyst has to decide which later revisions count. The first decision is the reverter population. Wikipedia’s quality work is done by bots, by trusted editors using one-click tools such as rollback, Huggle and Twinkle, by other registered and unregistered editors who use the undo function or edit by hand, and by authors who take back their own edits (Geiger & Ribes, 2010; Halfaker & Riedl, 2012; de Laat, 2015). The second is the attribution rule that links a later revision to the edit it undoes, on which the Wikimedia history dumps, the mwreverts library (Halfaker, n.d.) and the software’s own revert tag differ. The third is the observation window. Research on digital trace data has shown that such preprocessing decisions, often undocumented, can drive results (Howison et al., 2011; Hill & Shaw, 2014; Geiger & Halfaker, 2017; Sen et al., 2021), and measurement theory asks that an operationalization be justified against the construct it claims (Jacobs & Wallach, 2021). For reverts, the accuracy of detection has been examined (Ekstrand & Riedl, 2009; Flöck et al., 2012), and Geiger and Halfaker (2017) showed for reverts between bots that the operationalization changes the conclusion. That reverters are of different kinds is also established for the English edition (Halfaker & Riedl, 2012; Halfaker et al., 2013). Reviews of Wikipedia scholarship in this journal organize it by readership and content (Okoli et al., 2014; Mesgari et al., 2015); how its commonest behavioral trace is operationalized, and how far it moves with the three decisions across editions and rules, has not been examined.

This study asks how the three decisions are handled in published work; how far, with the measured edits held fixed, the newcomer revert rate, editor rankings and the association between being reverted and staying move when the decisions change; and whether reverts by different classes of reverter are the same kind of event. The first question is answered with a structured review of practice, the others with a fixed-edit audit of 3.2 million newcomer first edits in seven language editions, in the spirit of a multiverse or specification-curve analysis (Steegen et al., 2016; Simonsohn et al., 2020).

The contribution is a specification of what this indicator of quality work indicates: the share of reverts made by each class of reverter in each edition and the later activity of the newcomers each class reverts; the extent of a documented behavior of the software that has not, to my knowledge, been measured (Endnote 2), namely that its revert tag records a revert only when the reverting edit is trusted or patrolled; and the dependence of window results on the attribution rule. The paper ends with proposed reporting items tied to the construct an analyst intends to measure.

Conceptual background

One trace, four constructs

If quality is relative to activity and context (Stvilia et al., 2007; Mai, 2013), a trace produced by quality work is not itself a quality score. Stvilia et al. (2008) describe information quality work in Wikipedia as a division of labor among agents in different roles. A revert is the trace of one act in that work, the removal of a contribution, and the constructs for which it is used read that act from four positions: its object (was the edit acceptable), the relation between the parties (are they in dispute), its recipient (what the contributor experienced) and its performer (who did the work). I assigned each study in the review reported in the next section to the construct named by its research question or by the variable the revert enters (Table 1); four studies that evaluate detection methods or report descriptive rates were left unassigned. The reverter population each construct calls for is my reading, not a finding of the review. As a *quality judgment on the edit*, every revert that removes the edit counts, whoever made it, though a self-revert is a different judgment from a rejection by others. As *conflict* between editors or over an article (Kittur et al., 2007; Yasseri et al., 2012), reverts by bots and reverts of vandalism are not disputes, and mutual or repeated reverts matter more than single ones. As *the contributor's experience of rejection*, in

research on newcomer socialization and retention (Choi et al., 2010; Halfaker et al., 2013; Morgan & Halfaker, 2018), self-reverts should not count and the mode of rejection matters. As *human quality-work effort*, in studies of how communities and algorithms share governance (Müller-Birn et al., 2013; Zheng et al., 2019), bot reverts are a separate category and the status of one-click tools has to be decided, because a tool revert is a human judgment delivered with machine assistance (Halfaker & Riedl, 2012).

Table 1. Constructs a revert is used to indicate in 41 reviewed studies, the reverter population each calls for (the author's reading), and how many studies using the construct give no indication of the relevant choices under the conservative tally defined in the next section (strict tally in parentheses). Four studies are unassigned. Quality judgment covers the quality of an edit or, as in Stvilia et al. (2007), of an article.

Construct	Studies	Reverter population called for	No indication on bot, tool or self-reverts	No observation window
Quality judgment on the edit	11	All reverters; self-reverts reported separately	3 (6)	6 (8)
Conflict	15	Human reverters other than the author; vandalism reverts set apart	1 (6)	14 (15)
Contributor's experience of rejection	9	Others only; mode (bot, tool, manual) distinguished	2 (3)	5 (7)
Human quality-work effort	2	Humans; bots separate; tools decided explicitly	0 (0)	1 (2)

Detecting and attributing reverts

The identity revert, an edit that returns a page to a byte-identical earlier state, is the standard operationalization (Kittur et al., 2007). Two implementations dominate. The mwreverts library searches a window of 15 revisions by default, a value its source attributes to Priedhorsky et al. (2007). The mediawiki_history dataset (Wikimedia Foundation, 2026) instead marks a revision as reverted if any later revision of the page restores a state that preceded it, with no radius and no time limit. A third source is the mw-reverted change tag, enabled on all Wikimedia wikis on 15 September 2020 (Endnote 2), which the software applies when it recognizes the reverting edit as an undo, a rollback or a manual revert within 15 revisions. An edit made through the undo function is recorded as a revert whether or not its result is identical to an earlier revision, so the tag and the hash-based rules are not nested.

The rules differ in what they attribute. Under the unbounded rule, an edit removed once can be "reverted" again years later by any revision that returns the page to a still earlier state.

Automation and edition differences

Language editions differ in institutions that shape reverting: some hold edits by new users for review through FlaggedRevs (Tran et al., 2022), some patrol recent changes, the rollback right is granted on different terms, and norms of interaction differ (Hara et al., 2010). Nor does discarding a bot's reverts after the fact show what humans would have done without the bot, a question that needs evidence such as the bot outage studied by Geiger and Halfaker (2013).

How reverts are operationalized in published work

Procedure

I examined empirical studies and official Wikimedia analyses from 2005 to 2026 in which a revert measure computed from revision histories is an outcome, an exposure, a label or a conflict indicator in the main analysis, or is itself the object of description. Studies in which reverting is one category in a coding of editing activity, or only a control variable, are listed in Table S1 and not tallied (four studies, among them Arazy et al., 2019, in this journal). A first search stage followed the citation networks of the methodological works cited above and keyword searches of OpenAlex, and yielded 34 studies. Because it favored computing venues, a second searched OpenAlex for information science and neighboring journals (1,169 records, 79 assessed) and screened the full text of 242 articles in this journal and the ASIS&T proceedings that mention Wikipedia and reverting; it added seven studies, three of them from this journal and the Journal of Documentation. Nine candidates, none in an information science journal, could not be read (Table S1). The set is a structured sample of 41 studies (25 published before 2019, 16 since; 22 on the English edition only), not a census and not a systematic review.

Each study was read for its detection rule, its treatment of reverts made by bots, by one-click tools and by the author, its observation window, and any sensitivity analysis over these choices. A language model (Claude) retrieved texts, extracted candidate passages and proposed codes. I checked the codes of every study against the source text, for four studies with the assistance of a second model (ChatGPT), and the released coding table gives the passage behind each code so that readers can check it. Two tallies are reported. The *strict* tally counts a choice as stated only if the article or its supplement states it for the main analysis. The *conservative* tally, used in the text, also counts a choice as recoverable when it is fixed by a cited procedure, a named library default, the named data source, released code or the design of the measure (the wording

"eventually reverted", a measure built on mutual reverts between two editors, the removal of bot accounts from the sample). It gives the smallest defensible number of studies in which a choice is unknown. No second human coder took part and no reliability coefficient is claimed for the final codes (an earlier model-to-model comparison is described in Table S1); the counts are one reader's judgments about absence.

Findings

Table 2. Reporting of revert operationalization in 41 studies, 2005 to 2026. Recoverable only: not stated in the article, but fixed by a cited procedure, a library default, the data source, released code or the design of the measure.

Choice	Stated in the article	Recoverable only	Neither
Detection rule	29	3	9
Search radius, in the 22 studies using an identity rule	1	4	17
Reverts made by bots	9 (modeled separately 5; excluded 2; counted 2)	10 (bot accounts removed from the sample)	22
Reverts made with one-click tools	5 (counted 3; modeled separately 2)	0	36
Self-reverts	16 (excluded, 6 of them by definition)	5	20
Observation window	5 (time-bounded 4; revision-bounded 1)	8	28
Sensitivity analysis over any revert choice	8		33
Sensitivity analysis over the reverter population	0		41

For nine studies no detection rule can be established, and of the 22 that use an identity rule the search radius is known for five (Table 2). Twenty-two give no indication of how bots are handled. Ten more remove bot accounts or bot-authored edits from the sample and are silent on reverts that bots perform, so the treatment of bot reverts is stated in nine: five treat the type of reverter as a variable or focus on it, two exclude bot reverts (West et al., 2010; Tsvetkova et al., 2016) and two count them explicitly. Thirty-six say nothing about one-click tools. An observation window is stated in five studies and recoverable in eight; for 28 it is unknown (for conflict measures computed over whole article histories it is in effect the extent of the dump). Nine studies give no indication on bots, tools or self-reverts (18 under the strict tally), one states all three (Halfaker et al., 2013), no study reports all the choices together, and none varies the reverter population as a sensitivity analysis. None of the three studies from information science journals states the treatment of bot reverts or an observation window.

Two features of this record shape the audit. First, the exclusions actually made are of self-reverts, commonly, and of bot reverts, rarely; no study excludes tool-assisted reverts. The audit therefore sizes every reverter class and treats tool exclusion as hypothetical. Second, one study builds a conflict indicator partly on the software's revert tags, for the Hebrew edition (Ariel, 2026), and none notes the conditions under which the mw-reverted tag is applied.

Method

Data and cohort

The data are the mediawiki_history dumps of version 2026-09 (Wikimedia Foundation, 2026). Seven editions were processed: English from January 2018 to September 2026 (reverts to states older than 2018 are not detected, which against the dumps' own flag affects 28 of 42,042 sampled English first edits from October 2020) and the complete histories of Italian, Korean, German, Polish, French and Japanese.

The unit is a registered editor whose first edit in any namespace was made in the article namespace; that edit is the measured edit. Editors whose first edit was elsewhere (6.5 to 13.1 percent of first-time editors) are outside the cohort, as are unregistered and temporary accounts and accounts flagged as bots. The primary cohort is self-created accounts whose first edit dates from 1 January 2019 to 31 March 2026, leaving at least 180 days of follow-up: 3,246,958 editors (Table 3). One edit per editor makes the editor the unit of inference. First edits that create a page can be removed only by deletion; their share is reported (Table S2).

Attribution rules

Three rules are compared. The *dump rule* is the revision_is_identity_reverted flag with the identifier of and delay to the first reverting revision. The refinery source shows that the flag groups a page's revisions by content hash over its entire history; the first occurrence of a hash is a base, later occurrences are reverts to that base, and a revision is reverted if a later revert has a base that precedes it (Endnote 1). I reimplemented the rule from the hashes on all Korean first edits and on year-stratified samples of 60,000 first edits in each other edition; status agreed with the flag on at least 99.9 percent of edits in every edition. The *radius rule* applies the mwreverts convention of 15 revisions to the same recomputation. The *tag rule* is the mw-reverted tag on the measured edit.

Reverter classes

The first revert of each measured edit is assigned to one class, in this order: *bot* (the dump's bot flags, current or historical); *one-click tool* (the reverting revision carries the mw-rollback, huggle or twinkle change tag); *self* (the reverting account is the author); *unregistered* (no account, including temporary accounts where an edition has introduced them); and *other manual* (any other registered account, including those using the undo function). The tool class is defined by tags, not by a common institution: the rollback right is appointed in some editions, is bundled with the automatically granted editor group of FlaggedRevs in German and Polish, and in Japanese was held by administrators only until a rollbacker group was added in 2020 (Endnote 2). Tag availability differs: rollback is tagged in all seven editions throughout; Huggle is tagged in English and French and Twinkle only in English, from September 2020 (Table S9). In five editions the tool class is therefore the exercise of the rollback right, while in English it also contains Twinkle, which any autoconfirmed account can use; tags of other tools are counted in Table S3. Class shares are reported with the number of reverting accounts and with intervals that resample those accounts.

Quantities

For a window W, p(W) is the share of measured edits reverted within W. Excluding a class in the *strict* sense recodes edits first reverted by that class as not reverted; the change in p equals the class's share of first reverts times p, an identity. The *replace* variant, the object of the registered tests, instead looks for the first later revert by a non-excluded reverter. To learn what such later reverts are, I record for every first revert the next manual revert of the same edit by a registered third party (an account that is neither the author nor the first reverter), by class of the first reverter and under both rules, with the clock started at the first revert, and compare bot and tool first reverts with manual first reverts by registered editors.

Downstream quantities are the share of editors who edit again between day 2 and day 60, by class of first revert within 24 hours, with accounts blocked within 60 days separated; the rank agreement of editor-level success rates (one minus the reverted share of an editor's first-90-day article edits, for editors with at least ten) with and without tool and bot reverts; and tag coverage, the share of reverts found by the radius rule whose measured edit carries no mw-reverted tag, for first edits from October 2020, by class of reverter, by whether the reverting account held a user

group carrying the autopatrol right at the time, and by the edition's patrolling regime (Endnote 2). The cohorts are complete populations, so intervals express the sampling of page histories and the sensitivity of an estimate to individual editors and reverting accounts, not inference to other editions or periods. They are Wilson intervals for shares and percentile intervals from 2,000 bootstrap resamples of editors for ratios; because reverts cluster in reverting accounts, which Wilson intervals ignore, key quantities are recomputed with resampling of those accounts (Table S10), and Table 4 reports those wider intervals. Quantities that need page histories use the 60,000-edit samples (all 47,629 Korean first edits with a usable history). Cells with fewer than 50 edits are not reported, as registered.

Registration and its limits

Design, hypotheses and code were registered on Zenodo (https://doi.org/10.5281/zenodo.23720117) after English, Korean, French and German had been analyzed and before the outputs for Japanese, Polish and Italian were opened. It is a hold-out replication with thresholds set from the exploratory editions, not a blind preregistration: the held-out editions had already been processed, and that their outputs were unopened is documented only by my own logs (Section S1 of the supplement, which lists all deviations). Four hypotheses were registered. H1: the replace effect of excluding bots and tools is larger within 24 hours than in the unbounded window (ratio R of unbounded to 24-hour effect below 1, with R of .75 or more read as most of the effect persisting). H2: the share of bot- or tool-reverted edits later reverted by a human within 24 hours is below .25. H3, a descriptive criterion: the tag and radius rules agree on at least 95 percent of edits, and more than half of the edits the dump rule marks reverted but the tag does not lie beyond the radius. H4: the odds ratio of retention for reverted versus non-reverted edits varies by a factor of at least 1.5 across nine filter-by-window cells. Decisions use one-sided 95 percent bootstrap bounds with Holm adjustment within edition. All four outcomes are reported; a registered negative control and a simulation were withdrawn as uninformative (Section S1). Every other analysis was designed after the first results and is exploratory in all seven editions; tables and figures are labeled accordingly.

Transparency

Code, aggregates, the review table, logs and the deviation log are public (https://github.com/beatsbywoni/wiki-revert-audit; archived release, Jeong, 2026). Editor-level

files are not released, a recorded departure from the registration, and usernames of reverting accounts are not retained in outputs. The study uses public edit histories, reports aggregates only and involved no interaction with editors; it was not submitted for ethics review on that ground. Generative AI (Claude, Anthropic; model versions and dated tasks are in the public AI assistance log) was used for three tasks: drafting analysis code, drafting and editing prose, and, in the review of practice, retrieving and screening texts, extracting passages and proposing codes, which I checked against the sources as described above. A second model (ChatGPT, OpenAI) assisted that check for four studies. Model agents were also used to critique drafts before submission. No model generated, altered or imputed the audit's data: every number of the audit is produced by the released deterministic code from unmodified dumps. The review codes, by contrast, were proposed by a model before I checked them, and the tallies are counts over the released coding table. I specified the design, made the analytic decisions, checked the code against the primary sources cited and against the independent recomputation reported above, and am responsible for every statement. The study received no funding, and I declare no competing interests.

Results

What a revert rate is made of

Within 24 hours, 14.0 percent (Japanese) to 37.0 percent (English) of newcomer first edits were reverted (Table 3, Figure 1). Outside English, 9 to 20 percent of first edits create a page and cannot be reverted in this sense (Table S2). The first revert came from a bot in 12.2 percent of cases in English and 4.2 percent in French, and in at most 0.3 percent elsewhere. One-click tools made 24.7 percent (Japanese) to 52.5 percent (Italian) of first reverts. Manual reverts by other registered editors made up 28.9 to 64.3 percent, 49 to 78 percent of them through the undo function; unregistered editors and the newcomers themselves made the rest.

Table 3. Cohorts and composition of first reverts within 24 hours (dump rule, first edits of self-created accounts, 2019 to March 2026). Shares are of first reverts; the strict effect of excluding a class, in percentage points, is its share times p. Tool accounts: accounts that made the tool reverts, with the share of the most active. Editions marked * were held out for registered tests. Composition is exploratory.

Edition	First edits	p(24 h)	Bot	One-click tool	Other manual	Unregistered	Self	Tool accounts (top share)
English	2,189,284	.370	.122	.448	.364	.023	.044	6,596 (.06)
Italian*	135,400	.342	.000	.525	.407	.024	.044	201 (.06)
Korean	47,633	.216	.003	.516	.289	.096	.096	93 (.19)

Edition	First edits	p(24 h)	Bot	One-click tool	Other manual	Unregistered	Self	Tool accounts (top share)
German	236,903	.256	.001	.410	.545	.017	.027	1,286 (.09)
Polish*	67,755	.222	.001	.315	.616	.030	.038	341 (.11)
French	346,938	.243	.042	.255	.643	.012	.047	316 (.12)
Japanese*	223,045	.140	.001	.247	.558	.080	.114	103 (.23)

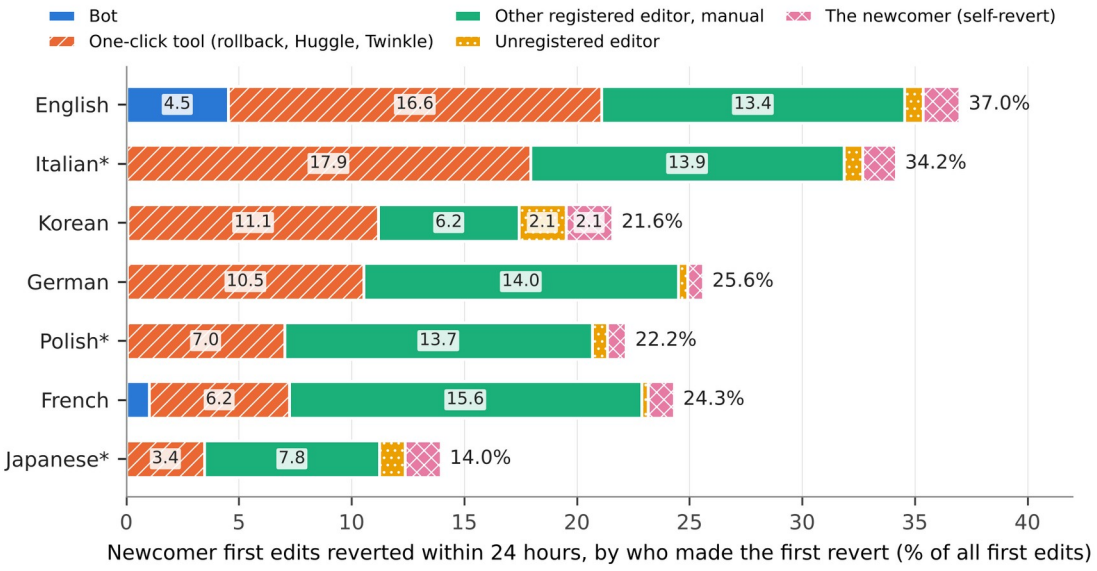


Figure 1. Newcomer first edits reverted within 24 hours, by who made the first revert, as a percentage of all first edits. Segment widths are the strict effects of excluding each class. Exploratory.

Excluding bot reverts lowers the 24-hour rate by 4.5 points in English and at most 1.0 elsewhere; excluding self-reverts by 0.7 to 2.1 points; excluding one-click tools by 3.4 (Japanese) to 17.9 points (Italian).

Three qualifications apply to the tool share. Outside English and German it is produced by few people: 93 accounts made the tool reverts in Korean and 103 in Japanese, the most active alone 19 and 23 percent, and intervals that resample reverting accounts are correspondingly wide (Table S2). In English the class is not constant over time, since Twinkle is tagged only from September 2020; the share is .524 for first edits from 2021. And it is a lower bound where other tools are tagged: reverts carrying only the tags of LiveRC in French, or of four further tools in English, would add 8.5 and 3.9 points (Table S3).

Later reverts and the window

If a class of reverts is excluded, does a later revert by someone else take its place? The registered tests addressed this for bots and tools together (Table S7). H2 was supported in the three held-out

editions: the share of bot- or tool-reverted first edits with a later non-bot, non-tool revert within 24 hours was .145 in Japanese, .111 in Polish and .114 in Italian, with upper bounds below the registered .25. Counting only registered third parties, and the 24 hours from the first revert, a manual revert followed 5.7 to 9.7 percent of bot and tool first reverts under the dump rule and 0.9 to 2.9 percent under the radius rule (Table 4, Figure 2).

Table 4. Follow-up manual reverts by a registered third party within 24 hours of the first revert, by who made the first revert, and the window ratio R, under two attribution rules. Manual: first revert by another registered editor without a tool. Denominators are all first reverts by the class. R: replace effect of excluding bots and tools, unbounded window over 24 hours, with 95 percent bootstrap intervals that resample reverting accounts. Follow-up shares are exploratory.

Edition	Dump rule: after bot or tool	Dump rule: after manual	Radius rule: after bot or tool	Radius rule: after manual	R, dump rule	R, radius rule
English	.077	.066	.010	.006	.92 [.90, .95]	1.05 [1.03, 1.07]
Italian*	.079	.082	.013	.010	.84 [.83, .86]	1.02 [1.02, 1.03]
Korean	.069	.068	.017	.010	.77 [.72, .84]	1.07 [1.05, 1.11]
German	.057	.041	.009	.003	.92 [.90, .94]	1.04 [1.02, 1.06]
Polish*	.060	.035	.013	.002	.88 [.85, .91]	1.07 [1.04, 1.10]
French	.090	.070	.013	.005	.86 [.84, .89]	1.04 [1.02, 1.06]
Japanese*	.097	.091	.029	.013	.92 [.89, .97]	1.05 [1.02, 1.10]

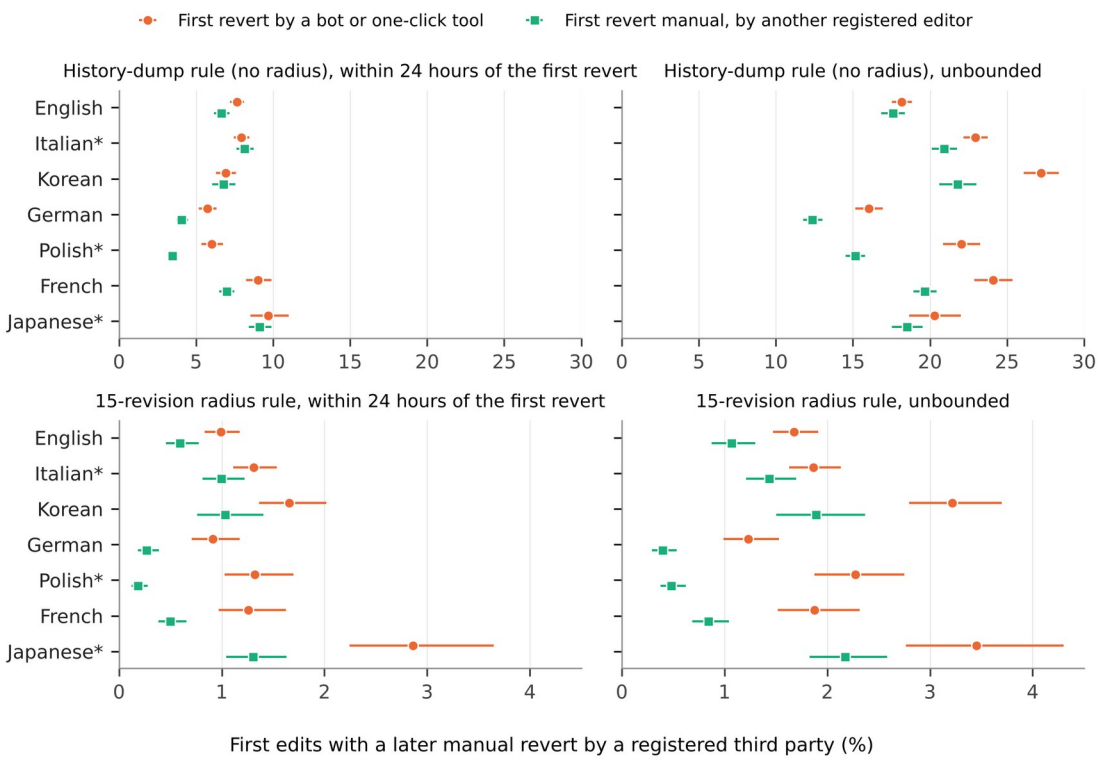


Figure 2. First edits with a later manual revert by a registered third party, by who made the first revert, under the dump rule (top) and the 15-revision radius rule (bottom), within 24 hours of the first revert (left) and unbounded (right). Bars are 95 percent Wilson intervals, which ignore clustering in reverting accounts. Exploratory.

By point estimate, such follow-up is somewhat more frequent after bot and tool reverts than after manual ones in six editions under the dump rule and in all seven under the radius rule (Table S4). What the data bound is the yield of the replace variant: an analyst who drops bot and tool reverts and looks for the next human revert recovers 6 to 10 percent of the dropped events within a day of the first revert (9 to 15 percent within a day of the edit under the registered definition), and under the dump rule mostly later reverts that reach back past the measured edit or undo unrelated edits; only 1.0 to 12.2 percent of bot- or tool-reverted first edits were reverted again after their content had been reinstated (Table S4).

H1 was supported under the dump rule in all three held-out editions (R of .92, .88 and .84, below 1; by the registered reading of values above .75, most of the effect persists; Table 4). Under the radius rule R is 1.02 to 1.07 in all seven editions, with every interval above 1. The reversal is close to arithmetic: a longer window adds first reverts by bots and tools and adds follow-up; under the radius rule follow-up is rare, so the effect grows, while under the dump rule deep reverts accumulate and the effect shrinks. H1 thus describes the unbounded rule more than it

describes reverting, and analysts who take the reverting revision from the dumps to ask who reverted inherit that rule.

Reverts by different classes reach different newcomers

Newcomers whose first edit was not reverted within 24 hours edited again on days 2 to 60 in 17.1 to 26.6 percent of cases (Figure 3). Of those reverted with a one-click tool, 3.5 to 8.6 percent did; of those reverted manually by another registered editor, 10.4 to 14.2 percent; of those who reverted themselves, 11.3 to 19.9 percent. As odds ratios against non-reverted newcomers, tool reverts range from 0.10 to 0.38, manual reverts by others from 0.41 to 0.71 and self-reverts from 0.55 to 0.68 (Table S5). Blocking does not produce the contrast: with accounts blocked within 60 days removed, the odds ratios are 0.09 to 0.38 for tool reverts and 0.40 to 0.70 for manual ones. These are associations between groups of newcomers, not effects of the mode of reverting: tools are aimed at vandalism, and intent, repeat accounts and accompanying warnings are not controlled. The point for measurement is that a single “reverted” category averages over events whose relation to the contributor’s later activity differs, by a factor of 1.4 to 4 in odds between manual and tool reverts, in proportions that vary by edition.

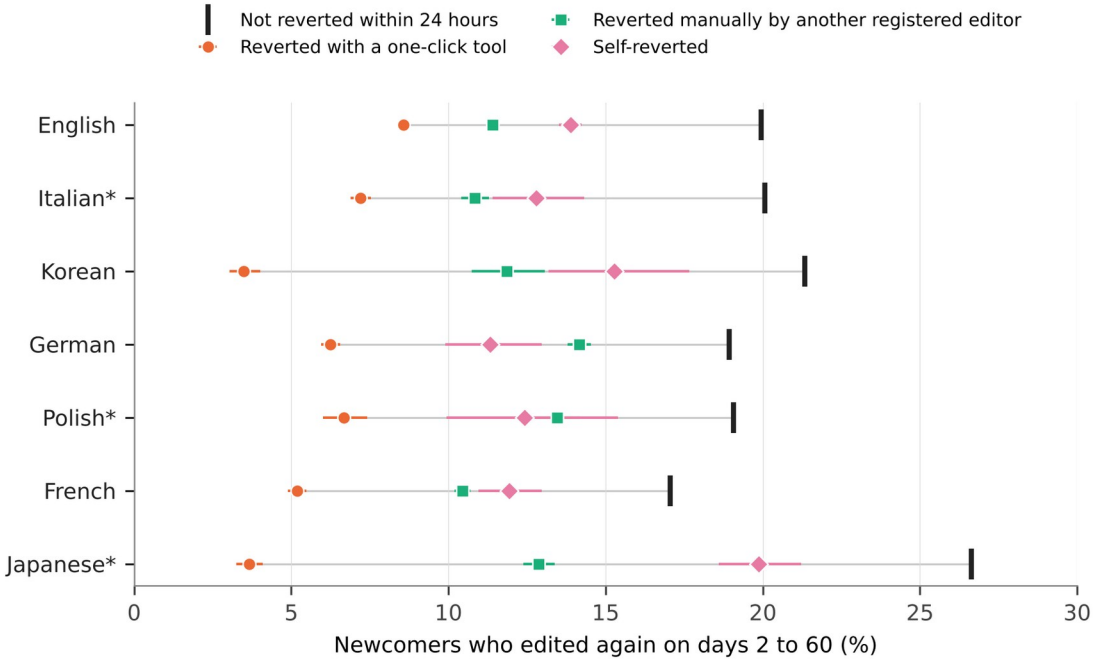


Figure 3. Newcomers who edited again on days 2 to 60, by who made the first revert of their first edit within 24 hours. Bars are 95 percent Wilson intervals; the grey line spans the classes within an edition. Exploratory.

This is what moves downstream estimates when the reverter population changes. The registered H4 compared the odds ratio of retention across nine cells (all reverts, without bots, without bots and tools; 24 hours, 7 days, unbounded). The ratio of the largest to the smallest was 1.84 [1.79, 1.89] in Japanese and 1.78 [1.72, 1.85] in Italian, above the registered 1.5, and 1.43 [1.37, 1.49] in Polish, below it; H4 is met in two of three held-out editions. The exploratory editions gave 1.51 to 1.78, and 3.01 in Korean. The movement is compositional: recoding tool-reverted newcomers, the group least likely to return, as not reverted raises the odds ratio. Across the three 24-hour cells alone, which do not overlap the outcome period, the ratio is 1.30 to 1.50 (1.93 in Korean). Editor-level rankings shift for the same reason: the rank correlation between editors' success rates with and without bot and tool reverts was .75 to .93.

The tag records a revert only when the reverter is trusted or patrolled

The tag and radius rules agreed on 98.6 percent of first edits in Japanese, 97.9 in Polish and 95.4 in Italian, meeting the first clause of H3. The second clause was rejected in all three: of the edits that the dump rule marked reverted and the tag did not, 42, 9 and 7 percent lay beyond the radius, against the registered expectation of more than half. From October 2020, after the tag was enabled everywhere, the share is 75 percent in Japanese, where the clause would be met (Table S10).

The exploratory follow-up locates the gap. MediaWiki applies the tag only once the reverting edit counts as approved: at once if the reverter holds the autopatrol right (exemption from patrol), and otherwise when someone patrols the edit, so that, in the words of the manual, "reverts that are not patrolled will not be marked with the reverted tag" (MediaWiki, n.d.; Endnote 2). For first edits from October 2020, the share of radius-rule reverts whose measured edit carries no tag was 0.3 to 1.0 percent in English, Korean and Japanese, where recent-changes patrolling is not enabled; 2.4 and 3.9 percent in German and Polish, which use FlaggedRevs; and 9.6 and 7.0 percent in Italian and French, which patrol recent changes (Figure 4). In Italian and French, 79 and 71 percent of reverts by unregistered editors and 71 and 73 percent of newcomers' self-reverts were untagged, against under 1 percent of tool reverts. In German and Polish the figures were 23 and 29 percent for unregistered reverters and 13 percent for self-reverts, which is consistent with approval by review under FlaggedRevs but was not measured. Because a later

approved revert by anyone also tags the edit, these class-specific shares are lower bounds on the omission per revert.

Patrol actions are not in the dumps, but the user groups held by the reverting account at the time of the revert test the mechanism indirectly (Table S6). Where the reverter held a group that carries the autopatrol right, 0.2 to 2.2 percent of reverts were untagged in every edition. Where a registered reverter held no group, 70 percent were untagged in Italian, 14 and 16 percent in German and Polish, and about 1 percent in English, Korean and Japanese (holders of global rights have no local group and fall in this row). French looked like an exception, with 12 percent, and resolves into the rule: that edition grants the autopatrolled status automatically at 500 edits and 90 days without recording a group (Endnote 2), and among manual reverters with no recorded group 60 to 70 percent of reverts were untagged below 500 edits, 3.4 percent between 500 and 4,999 and 0.1 percent above.

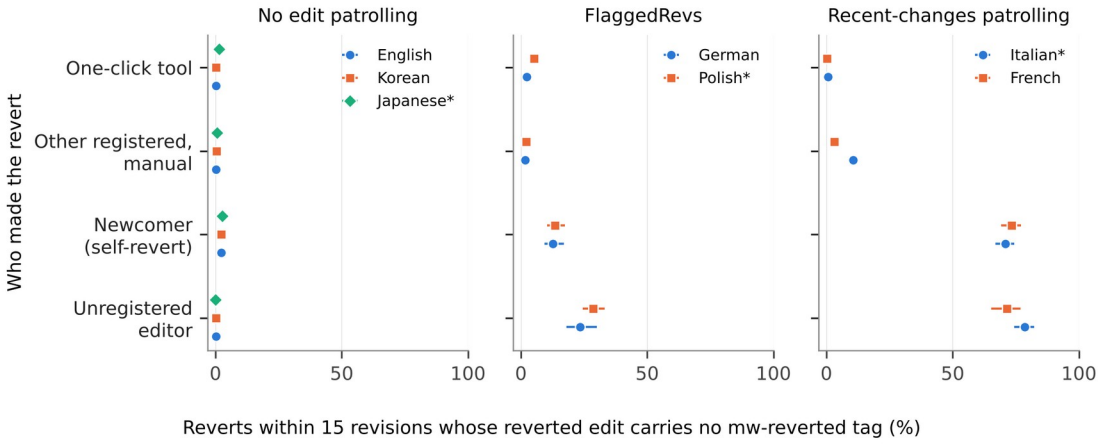


Figure 4. Reverts found by the 15-revision radius rule whose reverted edit carries no mw-reverted tag, first edits from October 2020, by who made the revert and by the edition's patrolling regime. Bars are 95 percent Wilson intervals. Exploratory.

The consequence for a comparison can be computed directly. For first edits from October 2020, the share reverted under the radius rule is 39.5 percent in English and 35.9 percent in Italian, a difference of 3.6 points; with the tag it is 40.1 and 32.7 percent, a difference of 7.4 points. The change of 3.8 points (95 percent interval 3.4 to 4.2 when reverting accounts are resampled) has two sources (Table S6). In Italian, 3.4 percent of first edits are reverted under the radius rule and untagged, and 0.3 percent are tagged without a radius-rule revert. In English the two figures are 0.1 and 0.7 percent: the tag also marks edits undone through the undo function with a result that

matches no earlier revision, and 292 of the 301 English edits tagged without a radius-rule revert are not identity-reverted under the dumps' own flag either (the truncated English history accounts for 6). The change is 3.7 points for first edits to 2024 and 4.0 from 2025, as the untagged share of Italian reverts rises from 9.0 to 12.5 percent. The French gap to English grows from 13.7 to 15.8 points, and the Korean and Japanese gaps do not change. The association of being reverted with retention is unaffected (odds ratios within .02 of each other under the two rules); what moves is any quantity that depends on who reverted. Neither rule is the truth, since the tag misses unapproved reverts and the hash-based rules miss undos that are not identical, and the example is a favorable case, a small gap between two large editions, measured on newcomer first edits only.

Discussion

What the indicator indicates

For each construct the audit shows what the unreported choices contain for newcomer first edits; among the edits of established editors, from which conflict measures draw most of their reverts, the shares will differ. As a *quality judgment on the edit*, the revert flag performs as intended only in its most inclusive form, and even then 3 to 11 percent of the judgments are the authors' own. As *conflict*, it needs its largest component set apart: a quarter to a half of first reverts are one-click responses, made with the rollback right or, in English, also with Huggle and Twinkle. As *the contributor's experience of rejection*, it mixes encounters that differ in what follows them, and the edition determines which is typical: a tool in Italian, Korean and English, a manual revert in German, Polish, French and Japanese. As *human quality-work effort*, it needs the least adjustment where it is most often adjusted: bots make a material share of first reverts only in English and French, while the status of one-click reverts concerns a quarter to a half of first reverts everywhere.

That bots, tools and other editors revert differently, and that tool reverts go with lower retention, was known for English (Halfaker & Riedl, 2012; Geiger & Halfaker, 2013; Halfaker et al., 2013), and that operational choices can reverse a conclusion about reverts was shown for bots (Geiger & Halfaker, 2017). The audit adds measurement: the size of each component in seven editions on one operational definition, the size and location of the tag's documented omissions,

and the dependence of timing results on the attribution rule. Neither rule is wrong: the unbounded rule answers whether content was ever removed, the radius rule who removed it.

Implications

For studies of quality work in the tradition of Stvilia et al. (2007, 2008), the consequence is to report the composition of reverts by reverter class with any rate and to choose exclusions from the construct.

The tag result is the most immediately usable. The mw-reverted tag is the cheapest revert indicator, and in the two editions that patrol recent changes it omits most reverts made by people without patrol-exempt status, who are disproportionately unregistered and new; under FlaggedRevs the omission is smaller, 13 to 29 percent for the same groups. Tag-based indicators are entering the literature: one reviewed study builds a conflict flag for the Hebrew edition from four revert tags and edit summaries (Ariel, 2026). Hebrew also enables recent-changes patrolling (Endnote 2), but its tag coverage was not measured here; it depends on how completely reverts by non-exempt editors are patrolled as well as on the configuration, and the three tags placed on reverting revisions are not subject to the condition. A comparison of tag-based revert rates across editions with different regimes can read a configuration difference as a difference in reception, as the English and Italian example shows, and the direction of the bias is known: reverts by the least established editors are the ones missing, and undos that are not identical are added.

Proposed reporting items

Table 5 separates what any study can state from what requires recomputation, in the manner of documentation standards with a low entry cost (Gebru et al., 2021). The items are a proposal from one audit, not a consensus guideline. Among the 41 reviewed studies, 29 state a detection rule (item 2), 5 a window (item 3) and one the treatment of all three reverter classes (item 4); the items completed for this study are in Table S11.

Table 5. Proposed reporting items for revert-based indicators.

Tier	Item
Always (no extra computation)	1. The construct the revert stands for (Table 1).
	2. The attribution rule: source field, library and radius, or tag.
	3. The observation window and its justification for the editions studied.
	4. Whether reverts by bots, by one-click tools and by the

Tier	Item
	author are counted.
When reverter data are available	5. The share of reverts made by each reverter class, and how many accounts make the tool reverts.
	6. Outcomes separately by reverter class when the construct is the contributor's experience.
When page histories are available	7. Results under both an unbounded and a radius rule if timing or the identity of the reverter matters.
For tag-based indicators	8. The edition's patrolling regime and autopatrol rules, the period of tag availability, and a coverage check against a hash-based rule.

Limitations

The measured edits are first edits of self-created accounts, so the audit says nothing directly about established editors or article-level conflict measures, where shares will differ. Reverter classes rest on bot flags and change tags: unflagged bots are few (reverts within 5 seconds by registered non-bot accounts are at most 0.15 percent of first reverts), but tools without a tag are counted as manual, and the English tags are incomplete before 2021. Edit filters, deletions and rejections of pending edits remove cases before they can be reverted, and partial reverts are missed by all hash-based rules. Intervals resample editors; resampling reverting accounts widens them without changing a decision (Table S10), and clustering in pages is not modeled. The registration cannot demonstrate that the held-out outputs were unseen, and the most informative analyses are exploratory and should be replicated on other editions and later dumps. Finally, the review is a structured sample read by one person with the assistance of language models, without a second human coder, and its inclusion criterion was narrowed after a first version; the coding table is released so that each code can be checked.

Conclusion

A revert is evidence of information quality work, and like other evidence it depends on who produced it and by what procedure it was recorded. A newcomer revert rate is a mixture of acts by bots, tool users, other editors and the newcomers themselves, in edition-specific proportions; the software's own record of reverts depends on the rights of the person who reverted, enough to change a difference between editions; and what a longer window shows depends on the attribution rule. Stating the construct, the reverter population and the rule costs a sentence each and makes revert-based findings comparable.

Data availability statement

The data that support the findings are openly available from the Wikimedia Foundation at https://dumps.wikimedia.org/other/mediawiki_history/ (version 2026-09 was used; only recent versions remain hosted). Code, aggregates, the review table, figure data, logs and the registration (https://doi.org/10.5281/zenodo.23720117; the registered text is also in the repository) are available at https://github.com/beatsbywoni/wiki-revert-audit and archived at Zenodo (https://doi.org/10.5281/zenodo.24054641). Editor-level derived data are not released because they could be re-identified from public edit histories; the code regenerates them from the dumps.

Endnotes

1. Column order and the identity-revert rule were read from the Wikimedia Analytics Refinery source (MediawikiEvent.scala; DenormalizedRevisionsBuilder.scala), master branch on 26 October 2026. A revision whose parent is the previous revision with the same hash is not counted as a revert, a filter added in commits of 28 May and 3 June 2026 (the source cites Phabricator task T266374); earlier dump versions may count such revisions.
2. Patrolling configuration was read from the Wikimedia configuration repository operations/mediawiki-config on 28 October and 12 November 2026: wgUseRCPatrol is enabled for French and Italian (and for Hebrew); FlaggedRevs is installed on German, Polish and English (on English for a small set of protected pages; the edition is grouped with those that do not patrol edits); Japanese and Korean have neither; French promotes accounts to autopatrolled implicitly at 500 edits and 90 days. In the same repository the editor group of German and Polish carries the rollback right, and the Japanese rollbacker group cites Phabricator task T258339. The mw-reverted tag was enabled on all Wikimedia wikis on 15 September 2020 (Phabricator task T164307). Its conditions are in the MediaWiki manual: approval of the reverting edit through autopatrol, patrol or FlaggedRevs review, and no tagging if the reverting edit has itself been reverted or deleted by the time it is approved. The maximum depth of 15 revisions is the default of the core setting $wgRevertedTagMaxDepth; that an undo counts as a revert without an exact match of content was read from the core source (EditResultBuilder.php, RevertedTagUpdate.php) on 12 November 2026. Task T266374 (2020) compares the tag

with the dumps' identity-revert field; I found no published measurement of tag coverage by reverter or edition.

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